

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. BPT-501: Mathematical Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

- (a) A conducting wire along the z -axis carries a steady current I . The resulting magnetic vector potential is given by:

$$\vec{A} = \hat{z} \frac{\mu_0 I}{2\pi} \ln \left(\frac{r_0}{r} \right)$$

Find the magnetic induction vector $\vec{B}$ in coordinates appropriate to the geometry.

- (b) The distance square ds^2 between two neighboring points in a 3D space is given in curvilinear coordinates by $ds^2 = g_{ij}dx^i dx^j$. Show that the metric coefficients g_{ij} transform as a covariant tensor of rank 2.

- (c) If A is a nonsingular symmetric matrix, show that A^{-1} is also symmetric.

- (d) From the Euler integral definition of the Gamma function $\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt$, show that:

(i) $\Gamma(z+1) = z\Gamma(z)$

- (ii) $\Gamma(z) = 2 \int_0^\infty e^{-t^2} t^{2z-1} dt$. State the condition imposed on z for the validity of this integral.

- (e) Find the Fourier series of the function $f(x) = |\sin(x)|$ in the interval $-\pi < x < \pi$. Comment on the nature of convergence of the series.

Question 2.

- (a) Normalize the Laguerre polynomial $L_2(x) = 1 - 2x + \frac{x^2}{2}$ in the range $[0, \infty)$ with weight function e^{-x} . Construct a second-order, ordinary, linear, homogeneous differential equation (2-ODE) that is satisfied by this polynomial. [6]

- (b) Verify the following relations in $\mathbb{R}^3$ using Levi-Civita symbols:

(i) $\epsilon_{ipq}\epsilon_{ipq} = 6$

(ii) $\epsilon_{ijk}\epsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}$

[8]

OR

- (a) Write down the spherical polar unit vectors in terms of Cartesian unit vectors. Using these, calculate all the partial derivatives of $\hat{r}, \hat{\theta}, \hat{\phi}$ with respect to r, θ, ϕ . [8]

- (b) Given a force vector in spherical polar coordinates:

$$\vec{F} = \frac{2P \cos \theta}{r^3} \hat{r} + \frac{P \sin \theta}{r^3} \hat{\theta}$$

express it in Cartesian coordinates. Does a potential function exist for this force? If so, find it. Provide an example of a physics problem in which the above expression occurs.
[6]

Question 3.

- (a) Show that the Bessel function $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ satisfies the Bessel differential equation:

$$x^2 y'' + xy' + \left(x^2 - \frac{1}{4}\right) y = 0$$

Find a second linearly independent solution of this equation. [7]

- (b) Given the generating function $g(x, t) = e^{2xt - t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$ for Hermite polynomials, obtain the Rodrigues representation:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2})$$

[7]

OR

- (a) Using the generating function $g(x, t) = (1 - 2xt + t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x) t^n$, ($|t| < 1$) for Legendre polynomials, show that:

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

[6]

- (b) Solve $(1 - x^2)y'' - 2xy' + 12y = 0$ by the Frobenius series solution method about $x = 0$, and obtain two independent solutions. [8]

Question 4.

- (a) By expressing the vector cross products in terms of the Levi-Civita symbols (ϵ_{ijk}), verify the vector identity:

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

[6]

- (b) Find the eigenvalues and normalized eigenvectors of the matrix A :

$$A = \begin{pmatrix} 4 & -1 & -1 \\ -1 & 4 & -1 \\ -1 & -1 & 4 \end{pmatrix}$$

How are these related to the eigenvalues and eigenvectors of A^2 ? [8]

Question 5.

- (a) Find the Fourier series expansion of the periodic function $f(x) = x^2$ in the interval $-\pi < x < \pi$. Comment on the nature of convergence of the series. Using this series, show that:

(i) $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

(ii) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$

[9]

- (b) Provide one example each, with a brief explanation, of physics problems in which the following special functions and distributions occur: Legendre polynomials, Bessel functions, Hermite polynomials, Laguerre polynomials, Gamma function, and Dirac delta function.

[5]